

Literature Review: Datasets for Knowledge Graph-Multimodal RAG in Medical Imaging

A survey of image, text, video, audio, and knowledge-graph resources relevant to building KG-grounded multimodal retrieval-augmented generation (KG-MMRAG) systems for medicine.

1. Introduction and Scope

Knowledge Graph-Multimodal RAG (KG-MMRAG) systems for medical imaging combine three ingredients: (i) large image-text corpora for cross-modal retrieval and grounding, (ii) structured knowledge graphs (KGs) that encode clinical/biomedical relations for reasoning and hallucination reduction, and (iii) evaluation benchmarks that stress-test retrieval and generation quality. Recent surveys frame this as a fast-moving area — the field's first dedicated multimodal RAG survey appeared only in February 2025, and subsequent taxonomies (Abootorabi et al. 2025; Zheng et al. 2025; Gao et al. 2025) have organized datasets, architectures, and training strategies for the broader multimodal RAG landscape, while medically-focused frameworks such as MKGF, MedRAG, and mKG-RAG apply KG-elicited reasoning specifically to clinical retrieval and VQA.

This review organizes the dataset landscape into eight categories: (2) large-scale image-text pretraining corpora, (3) medical visual question answering (VQA) / instruction datasets, (4) segmentation and classification imaging datasets, (5) knowledge graphs and ontologies, (6) clinical text and entity-relation extraction datasets, (7) video datasets, (8) audio datasets, followed by (9) RAG-specific medical benchmarks and (10) a synthesis table with notes relevant to KG-MMRAG pipeline design (retrieval corpus vs. KG substrate vs. evaluation set).

2. Large-Scale Image-Text Pretraining Corpora

These datasets are the backbone of the *dense retrieval* side of KG-MMRAG — they supply the (image, caption/report) pairs used to train or fine-tune vision-language retrievers (e.g., BiomedCLIP-style encoders) and the multimodal index that a RAG system searches over.

Dataset	Scale	Modality coverage	Notes
MIMIC-CXR / MIMIC-CXR- JPG	377,110 images, 227,835 studies, 65,379 patients	Chest X-ray + free-text radiology reports	De-identified, PhysioNet-gated; each report carries structured 14-label CheXpert-derived annotations; the largest widely-used report-paired CXR corpus and a de facto standard benchmark for report generation and retrieval.
IU-Xray (Open-I)	~7,470 images / ~3,955 reports	Chest X-ray + reports	Smaller, non-gated alternative to MIMIC-CXR; commonly paired with MIMIC-CXR in retrieval and report-generation studies (used in Nathan's KG-MMRAG pipeline).
ROCO / ROCOv2	ROCO: ~80K radiology + ~6K non-radiology images; ROCOv2 expands coverage	Multi-modality radiology images + captions/keywords, mined from PMC	One of the earliest general-purpose biomedical image-caption datasets; ROCOV2 improves label/caption quality and modality diversity, and is heavily underrepresented in ultrasound (~5% of images).
PMC-OA	~1.6-12.2M figure-caption pairs (2.48M source papers)	Mixed biomedical figures from PubMed Central Open Access	Automated caption-extraction pipeline; captions are less detailed than manual clinical reports, limiting fine-grained supervision.
PMC-15M	15M image-text pairs	Diverse biomedical image types	Two orders of magnitude larger than MIMIC-CXR; used to pretrain BiomedCLIP, a widely adopted biomedical multimodal foundation encoder (also used in Nathan's KG-MMRAG stack).
Open-PMC- 18M	18M image-text pairs	PMC-derived, high- fidelity re-curation	Positioned as a quality-first alternative/superset to PMC-15M and PMC-OA, addressing noise in earlier PMC-mined datasets.
BIOMEDICA	~24M pairs	PMC-derived, taxonomy-labeled	Assigns global/local modality labels via clustering and vision-encoder-based classification; currently among the largest released PMC-derived corpora.

Dataset	Scale	Modality coverage	Notes
MedTrinity-25M	25,016,845 samples across 10 modalities, 65+ diseases	Radiology, pathology, and more, with multigranular {image, ROI, description} triplets	First automated pipeline generating multigranular (global + local/ROI-level) annotations without requiring paired text, aggregated from TCIA, Kaggle, Zenodo, Synapse, Hugging Face, Grand Challenge, GitHub, CheXpert, DeepLesion and BraTS-family sources.
Quilt-1M	~1M (802K-1M reported) image-text pairs	Histopathology, mined from YouTube video frames/subtitles, Twitter, papers	Largest histopathology vision-language dataset at release; used to fine-tune CLIP-style models, outperforming prior SOTA on 13 patch-level pathology classification datasets.
CheXpert	~224K chest radiographs	Chest X-ray with uncertainty-aware structured labels	Companion/benchmark dataset to MIMIC-CXR from Stanford; frequently used as an out-of-distribution test set for models pretrained on MIMIC-CXR.
Harvard-FairVLMed	Fundus images + text	Ophthalmology	Focused on fairness evaluation of multimodal models across demographic subgroups — relevant if a KG-MMRAG system needs bias auditing.

3. Medical Visual Question Answering (VQA) and Instruction Datasets

These datasets are typically used to fine-tune/evaluate the *generation* side of a KG-MMRAG pipeline (i.e., how well the system answers grounded clinical questions), and several — notably SLAKE — ship with their own embedded knowledge graph, making them directly relevant to KG-RAG evaluation.

Dataset	Scale	Notes
VQA-RAD	315 images, 3,515 QA pairs (517 possible answers), 11 question categories	Clinician-authored, built from the MedPix database; covers head/chest/abdomen across X-ray, CT, MRI. The oldest manually-curated med-VQA benchmark still in wide use.
SLAKE	642 images, ~14,000 QA pairs (English + Chinese)	Covers 12 diseases, 39 organs, X-ray/CT/MRI; ships with semantic segmentation masks, bounding boxes, and an embedded medical knowledge graph of 5,232 triplets — arguably the most directly KG-relevant med-VQA dataset available.
PathVQA	4,998 pathology images, 32,799 QA pairs	Built via a semi-automated pipeline from pathology textbooks and a digital library; open- and closed-ended questions on location, shape, color, appearance.
PMC-VQA	~227K QA pairs, ~149K images	Large-scale, broad modality/disease coverage, multiple-choice format; generated to support generative med-VQA model development.
Quilt-VQA	344 closed + 957 open QA pairs	Pathology-specific VQA benchmark derived from the Quilt corpus.
OmniMedVQA	Multi-modality, multi-organ	Broadens VQA coverage across body regions/modalities to test generalization beyond radiology/pathology.
MIMIC-Diff-VQA / Medical-Diff-VQA	Derived from MIMIC-CXR	Emphasizes <i>differential</i> diagnosis reasoning between paired chest X-rays (e.g., current vs. prior study) — useful for testing temporal/relational reasoning in a KG-RAG setting.
GEMeX / GEMeX-RMCoT	Region-aware, chain-of-thought annotated	Adds both textual and visual explanations and region-grounded multimodal CoT reasoning to Med-VQA.
ProbMed	6,303 images, 57,132 QA pairs	Probing benchmark spanning modality recognition, organ recognition, clinical findings, and diagnostic reasoning dimensions.

4. Segmentation, Classification, and Structural Imaging Datasets

These datasets don't come with narrative text but are essential as the "image knowledge" side of a KG-MMRAG system (e.g., populating a KG with segmented anatomical/lesion entities, or supplying the visual backbone data used in Nathan's NEARCAM/BraTS-adjacent segmentation work).

Dataset	Scale	Notes
BraTS (2020 / 2021 / 2024 / MEN-RT)	BraTS 2020: 369 patients, 4 MRI sequences (T1, T1ce, T2, FLAIR); BraTS 2021: 1,251 scans; BraTS 2024 MEN-RT: 570-750 radiotherapy-planning MRIs	Longest-running annual MICCAI brain tumor segmentation challenge; multi-institutional, expert-annotated, co-registered/skull-stripped NIfTI volumes; recent editions extend to meningioma radiotherapy planning and post-treatment glioma.
DeepLesion	~32K lesions	CT lesion detection across body regions; incorporated into MedTrinity-25M's aggregation pipeline.
RadGenome-Chest CT	Chest CT with richer region-level annotations	Still dependent on paired image-text data, limiting scale relative to auto-annotated corpora like MedTrinity-25M.
VinDr-PCXR	Large-scale pediatric CXR	Open dataset for interpretation of common pediatric thoracic diseases.
DeepDRiD	Diabetic retinopathy	Grading + image-quality estimation challenge dataset.
BRSET	Brazilian multilabel ophthalmological dataset	Regional/demographic diversity for fundus imaging tasks.

(Nathan's own NEARCAM RGB-D tapping-line dataset and BraTS 2020 baselines used in his ISI Kolkata segmentation work fall in this category as domain-specific extensions of the same paradigm — structural/segmentation data with sparse or no accompanying narrative text, requiring separate KG-grounding strategies than caption-rich corpora.)

5. Medical/Biomedical Knowledge Graphs and Ontologies

This is the "KG" backbone category — structured relational resources that a KG-MMRAG system queries or traverses alongside dense retrieval. Sizes vary hugely depending on integration scope.

Knowledge Graph / Ontology	Entities	Relations/Edges	Notes
UMLS (Unified Medical Language System)	3.7M+ concepts from 200+ source vocabularies	—	Meta-thesaurus aggregating SNOMED CT, RxNorm, MeSH, and others; broad coverage but inconsistent cross-vocabulary structure.
SNOMED CT	~358,000 clinical concepts	SFull KG: 302,332 entities, 229 relation types, 4,129,726 triples	Gold-standard formal-logic clinical terminology; commonly extracted as a sub-KG from UMLS for KG-completion or entity-linking research.
PrimeKG	129,375 nodes (17,080 diseases)	4,050,249 relationships, 29–30 edge types	Integrates 20 biomedical resources (Mayo Clinic, Orphanet, DrugBank, MONDO, etc.) across ten biological scales; augmented with clinical-guideline text descriptions for multimodal analysis — one of the most-cited precision-medicine KGs for LLM/RAG grounding.
Hetionet	47,031 nodes	2,250,197 relationships, 24 relation types	Heterogeneous network integrating 29 sources (ChEMBL, BindingDB, etc.), predicting gene–disease associations across 29 complex diseases.
DRKG (Drug Repurposing Knowledge Graph)	97,238 entities	5,874,261 triples, 107 relation types	Relates genes, compounds, biological processes, side effects, and diseases; includes COVID-19-specific extensions.
PharmKG	7,601 entities	500,958 relations, 29 relation types	Focused on pharmacology (OMIM, PharmGKB-derived).
SPOKE	1,143,313 nodes	32,806,467 edges, 55 relation types	One of the largest integrated biomedical KGs (15+ source ontologies).
RTX-KG2	500,000+ nodes	30,000,000+ edges	Large aggregator KG used in NCATS Biomedical Data Translator infrastructure.
Clinical Knowledge	16M+ nodes	220M+ relationships	Built from 25+ databases via Neo4j; targeted at proteomics-driven

Knowledge Graph / Ontology	Entities	Relations/Edges	Notes
Graph (CKG)			precision medicine workflows.
iBKH	2M+ entities	4.8M+ relations	Integrative biomedical knowledge hub aggregating diverse resource types.
CTKG	1.5M+ nodes	3.6M+ triples	Derived specifically from ClinicalTrials.gov.

Relevance note: for a project like KG-MMRAG combining BiomedCLIP + MedGemma + FAISS + networkx/Neo4j, PrimeKG and RadGraph (below) are the two most directly applicable off-the-shelf KGs — PrimeKG for broad disease/drug/phenotype relational grounding, RadGraph for radiology-report-native entity/relation structure that maps directly onto MIMIC-CXR/IU-Xray images.

6. Clinical Text and Entity-Relation Extraction Datasets

These datasets bridge unstructured clinical text and structured KGs — i.e., they are the resources used to *build or validate* a domain-specific KG from radiology/clinical narrative text, which is central to constructing the graph side of a KG-MMRAG pipeline.

Dataset	Scale	Notes
RadGraph	Dev set: 500 MIMIC-CXR reports (14,579 entities, 10,889 relations); Test set: 100 reports (MIMIC-CXR + CheXpert); Inference set: 220,763 MIMIC-CXR reports (~6M entities, ~4M relations) + 500 CheXpert reports	Board-certified radiologist-annotated entity/relation schema for chest X-ray reports; each report can be linked back to its associated image, making it a natural bridge dataset for image-grounded KG construction.
RadGraph-XL	2,300 reports, 410,000+ entities/relations	Extends RadGraph beyond chest X-ray to chest CT, abdomen/pelvis CT, and brain MRI — multi-anatomy, multi-modality entity/relation extraction.
MedNLI	Derived from MIMIC-III clinical notes	Natural language inference dataset for clinical text, useful for training/evaluating textual entailment components in a RAG verifier.
MedQA (USMLE)	1,273 four-option test questions (English subset used in MIRAGE)	Multi-choice QA from professional medical board exams; a standard text-only reasoning benchmark, frequently paired with retrieval corpora (PubMed, textbooks, StatPearls, Wikipedia) in MedRAG-style systems.
MedMCQA	194K questions (4,183 in dev-set benchmarks)	Indian medical entrance exam questions across 21 subjects/2.4K topics.
PubMedQA	1K expert-annotated (500-question test split commonly used)	Yes/no/maybe biomedical research QA grounded in PubMed abstracts; explicitly designed to test evidence retrieval and reasoning.
BioASQ	Ongoing challenge series	Biomedical semantic QA/IR benchmark, frequently included alongside PubMedQA in medical RAG evaluation suites.

7. Video Datasets

Video is the least-represented modality in current KG-MMRAG literature but is increasingly used for surgical/procedural multimodal reasoning.

Dataset	Scale	Notes
Cholec80	80 videos, ~370,168 frames, 25 FPS, ~51 hours raw	Laparoscopic cholecystectomy; annotated with 7 surgical phases and 7 tool-presence labels; the most widely used surgical video benchmark, underlying numerous derivative datasets.
CholecT45 / CholecT50	45–50 videos derived from Cholec80/Cholec120	Fine-grained ⟨instrument, verb, target⟩ action-triplet annotations plus phase labels — closest existing analog to a "video knowledge graph" for surgery.
CholecSeg8k	8,080 frames	Pixel-level segmentation masks for 13 anatomical/instrument classes, derived from Cholec80.
Cholec80-CVS	Same 80 videos	Adds Critical View of Safety (CVS) criteria annotations for bile-duct-injury risk assessment.
Cataract-101	101 cataract surgery videos	11 phase labels; used alongside Cholec80 in cross-procedure multimodal video-language pretraining (e.g., SurgVLP).
HeiChole	Endoscopic surgical videos	Used alongside Cholec80 for surgical report/documentation generation research.
Cholec80-VQA / SSG-VQA	Derived VQA sets	Early attempts converting single-task surgical annotations (phase/tool recognition) into VQA format; limited annotation depth motivated newer aggregated benchmarks.
SurgVLM-DB / EndoBench / Surg-3M	Large aggregated multi-source	Recent (2025) large-scale surgical vision-language corpora combining multiple procedures/organs for instruction tuning, analogous in spirit to PMC-OA/PMC-15M but for procedural video.

8. Audio Datasets

Audio is used for two distinct purposes in medical multimodal systems: (a) diagnostic auscultation sound classification, and (b) doctor-patient conversational/dictation understanding — the latter being directly relevant to voice-interactive RAG interfaces (e.g., Nathan's ZeroGPU voice-in/voice-out crisis-triage demo is architecturally similar to what a voice-driven KG-MMRAG front end would need).

Dataset	Scale	Notes
ICBHI 2017 Respiratory Sound Database	920 recordings, 126 subjects, 5.5 hours, 6,898 annotated respiratory cycles	The standard benchmark for lung-sound (crackle/wheeze) classification; recorded with 4 stethoscope types across variable sampling rates.
KAUH	112 subjects	Respiratory sounds with diagnosis, sound-type, and chest-zone metadata.
CirCor	1,568 subjects (pediatric)	Heart sound recordings with murmur and segmentation annotations.
SPRSound	292 participants (pediatric)	Record- and event-level respiratory sound annotations.
ZCHSound	1,259 participants (pediatric), 566 congenital heart disease cases	Pediatric heart sound database.
AusculataBase-Corpus	11 aggregated datasets, 40,317 recordings, 322.4 hours	Unified heart/lung/bowel sound corpus with a 16-task benchmark suite spanning binary, multi-class, multi-label, and regression tasks — the most comprehensive auscultation aggregation to date.
HLS-CMDS	535 recordings	Manikin-recorded heart/lung/mixed sounds via digital stethoscope, useful for controlled/synthetic augmentation.
CaReSound	Aggregated from ICBHI, KAUH, CirCor, SPRSound, ZCHSound	Purpose-built for a Cardiac/Respiratory Audio Question Answering task — i.e., an audio-native analog of Med-VQA, directly relevant to extending KG-MMRAG to the audio modality.
Fareez et al. OSCE respiratory dataset	272 audio files (MP3) + transcripts	Synthetic patient-physician interviews structured as Objective Structured Clinical Examinations, targeting respiratory cases.
Kaggle Medical Speech dataset	8.5 hours audio-text pairs	Common medical symptom utterances paired with transcriptions; used for medical ASR adaptation.

9. RAG-Specific Medical Benchmarks (Retrieval Evaluation)

These are not raw content corpora but *evaluation harnesses* for RAG systems themselves —

directly relevant to validating a KG-MMRAG pipeline's Recall@k, nDCG, and hallucination metrics (mirroring Nathan's own KG-MMRAG evaluation harness design).

Benchmark	Scale	Notes
MIRAGE	7,663 questions across 5 QA datasets (MMLU-Med, MedQA-US, MedMCQA, PubMedQA*, BioASQ-Y/N)	First benchmark purpose-built to evaluate medical RAG systems; revealed log-linear retrieval-scaling and "lost-in-the-middle" effects specific to medical RAG.
MultiMedQA	6 combined QA datasets + consumer-query set	Broader multi-task medical QA benchmark (used in Med-PaLM evaluation), includes a human evaluation framework for answer quality.
MedRGB	Built on 4 MIRAGE-derived QA sets	Explicitly tests retrieval-augmented generation robustness (not just accuracy) for medical QA.
RAGCare-QA	420 theoretical medical knowledge questions	Focused on evaluating RAG pipelines in medical <i>education</i> contexts rather than clinical decision support.
MMed-RAG evaluation suite	5 datasets: MIMIC-CXR, IU-Xray, Harvard-FairVLMed, PMC-OA, Quilt-1M	Directly evaluates a <i>multimodal</i> medical RAG system spanning radiology, fundus imaging, and pathology — closest existing published analog to a full KG-MMRAG evaluation protocol.
MedRAG (KG-elicited reasoning benchmark)	Manifestation → diagnosis/treatment retrieval task	Evaluates KG-grounded reasoning specifically, pairing retrieved KG triples with generation quality.

10. Synthesis: Mapping Datasets to a KG-MMRAG Pipeline

Pipeline component	Best-fit dataset categories	Representative choices
Dense multimodal retriever training/index	Image-text pretraining corpora (§2)	PMC-15M / PMC-OA / MedTrinity-25M / Quilt-1M / MIMIC-CXR

Pipeline component	Best-fit dataset categories	Representative choices
KG substrate (structured reasoning)	Knowledge graphs & ontologies (§5)	PrimeKG (broad clinical), RadGraph (image-native radiology)
KG construction/validation from text	Clinical text & entity-relation extraction (§6)	RadGraph, RadGraph-XL, MedNLI
Generation fine-tuning / instruction data	Med-VQA & instruction datasets (§3)	SLAKE (has its own KG), PMC-VQA, PathVQA
Segmentation-grounded visual entities	Structural imaging datasets (§4)	BraTS family, DeepLesion, MedTrinity-25M ROI annotations
Video-modality extension	Surgical video datasets (§7)	Cholec80/CholecT50 family for procedural KG-RAG
Audio-modality extension	Auscultation & conversational audio (§8)	CaReSound (audio QA-native), AusculBase
End-to-end evaluation	RAG-specific benchmarks (§9)	MIRAGE, MMed-RAG suite, MedRGB

Observed gaps in the current literature (as of mid-2026):

1. **Ultrasound is systematically underrepresented** across nearly every major image-text corpus (well under 5% of samples in PMC-15M, PMC-OA, MedTrinity-25M, and ROCO/ROCOv2), which is a concrete opportunity area.
2. **Video-text and audio-text pairing at PMC-15M/MedTrinity-25M scale does not yet exist** — surgical video and auscultation audio datasets remain orders of magnitude smaller than the image-text corpora, and are largely classification-labeled rather than caption/report-paired, limiting direct reuse for dense video/audio retrieval.
3. **Very few datasets combine an explicit KG with paired imaging and free text simultaneously** — SLAKE is the clearest exception (image + QA + 5,232-triplet KG); most KG resources (PrimeKG, Hetionet, SNOMED CT) are image-agnostic and must be linked to imaging data via a separate entity-linking step (e.g., via RadGraph as a bridge for chest imaging), which is exactly the kind of integration challenge a KG-MMRAG system needs to solve architecturally rather than inherit "for free" from a single dataset.
4. **Cross-modal RAG evaluation is still nascent:** MIRAGE and MultiMedQA evaluate text-only medical RAG; MMed-RAG is one of very few benchmarks evaluating a genuinely *multimodal* medical RAG system end-to-end, and none of the current benchmarks span image + video + audio + KG jointly.

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Compiled July 2026. This review reflects the published literature and dataset releases available at the time of writing; several 2026 preprints (e.g., MedGPT-oss, Open-PMC-18M follow-ups) indicate the corpus landscape is still actively expanding, particularly around quality-filtered PMC-derived pretraining sets.